

College Management System

Faculty Module Documentation

Comprehensive ERP User Manual, Technical Specification & System Architecture

Document Type:	System Specification & User Manual
Module:	Faculty Module (Enterprise ERP)
Version:	Version 1.0
Prepared By:	Development Team
Academic Year:	2026-27
Institution:	EduSuite Pro College ERP
Date Generated:	August 6, 2026
Status:	Approved & Backend-Ready

Table of Contents

1. Introduction	Page 3
2. Faculty Workflow	Page 3
3. Dashboard	Page 4
4. My Profile	Page 4
5. Timetable	Page 5
6. Subjects	Page 5
7. Lesson Plans	Page 6
8. Attendance	Page 6
9. Students	Page 7
10. Assignments	Page 7
11. Study Materials	Page 8
12. Assessments	Page 8
13. Examinations	Page 9
14. Research	Page 9
15. Leave	Page 10
16. Payroll	Page 10
17. Reports	Page 11
18. Notifications	Page 11
19. Settings	Page 12
20. Role-Based Access Control	Page 12
21. Future Backend APIs Specification	Page 13
22. Database Entities & Schema Models	Page 14
23. UI Screens Specification	Page 15
24. Technology Stack	Page 16
25. Security & Authentication Architecture	Page 16
26. Responsive Design & Accessibility	Page 17
27. Future Enhancements & AI Integration	Page 17
28. Conclusion & System Summary	Page 18

1. Introduction

Overview, Purpose, Objectives & Scope of Faculty Module

The Faculty Module of the EduSuite Pro College Management System (CMS) is an enterprise-grade ERP workspace tailored to empower academic faculty, heads of department (HODs), and academic managers in managing their day-to-day instructional, administrative, evaluation, and research responsibilities.

Purpose

The primary purpose of the Faculty Module is to eliminate paper-based administrative friction, standardize course delivery tracking, provide real-time attendance & assessment entry, and present centralized analytics for institutional decision making.

Key Objectives & Benefits

- **Unified Workflow:** Consolidate teaching schedules, attendance marking, syllabus tracking, and student evaluations into one responsive web interface.
- **Department Awareness:** Automatic context filtering tailored to Computer Science, Mechanical, ECE, Civil, and Management departments.
- **Master Timetable Synchronization:** Single source of truth for institutional schedules with zero duplicate mock data maintenance.
- **Continuous Internal Evaluation (CIE):** Seamless marks entry for internal assessments, lab vivas, assignments, and semester end exams.
- **Accreditation Readiness:** Out-of-the-box generation of NBA Outcome-Based Education (OBE) and NAAC Criterion 2, 3 & 5 compliance reports.

2. Faculty Workflow

End-to-End Operational Lifecycle

The diagram below illustrates the sequential workflow executed by faculty members during their daily operational lifecycle inside the College Management System:

Faculty Operational Lifecycle Flow

- [1. Login] % % % % % % % % % %° Secure SSO / Credentials & Multi-Factor Auth
- [2. Dashboard] % % % % % % % %° Executive KPI Overview, AI Insights, Today's Classes
- [3. Timetable] % % % % % % % %° Master Schedule Sync & Classroom Allocation
- [4. Attendance] % % % % % % % %° Real-Time Section Roster Marking & Defaulter Alerts
- [5. Lesson Plans] % % % % % %° Session Planner, Covered Topics & Progress Tracker
- [6. Materials & LMS] % % %° Upload Lecture PDFs, PPTs & Video Resources
- [7. Assignments] % % % % % %° Create Tasks, Evaluate Submissions & Feedback
- [8. Assessments] % % % % % %° Internal Marks, Quiz, Viva & Project Grading
- [9. Examinations] % % % % % %° Invigilation Duties & Question Paper Uploads
- [10. Research] % % % % % % % %° Publications, Grants, Patents & Profile Export
- [11. Leave & Payroll] % %° Apply Leave, Track Balances & View Payslips
- [12. Reports] % % % % % % % %° Export Attendance, Marks, NBA & NAAC Compliance
- [13. Notifications] % % %° Instant Alerts for Deadlines & Student Requests
- [14. Settings] % % % % % % % %° Profile Edit, Security Passwords & Preferences

3. Dashboard

Centralized Command Center & Analytics Widgets

The Faculty Dashboard serves as the command center for professors and department heads, displaying live widgets, course summaries, upcoming lecture countdowns, and automated AI academic insights.

Core Dashboard Components

- **Today's Schedule Widget:** Displays real-time breakdown of current day lectures, assigned room numbers, student strength, and quick navigation.
- **AI Academic Insights:** Automated anomaly detection highlighting low attendance classes, pending assignment evaluations, and syllabus delays.
- **KPI Summary Cards:** Key metrics displaying Total Teaching Hours, Pending Marks Entry, Average Attendance %, and Active Research Grants.
- **Quick Action Shortcuts:** One-click shortcuts to Mark Attendance, Upload Syllabus, Post Announcements, and Submit Internal Marks.
- **Role-Based Adaptability:** HODs automatically view department-wide workloads and leave approvals; Assistant Professors view personal course rosters.

4. My Profile

Academic Credentials & Contact Directory

The Profile module enables faculty to manage their official institutional CV, academic designations, office hours, and cabin consultation slots.

Profile Field	Field Description	Access / Visibility
Faculty Name & ID	Official Full Name & Employee Identifier (EMP-CSE-2041)	Public Directory
Academic Designation	Professor, Associate Professor, HOD	Institutional Roster
Department	Computer Science & Engineering (CSE)	Department Wide
Cabin & Office Room	Block A, Room 302 / Innovation Center Lab 5	Student Visible
Counseling Hours	Mon, Wed 03:30 PM - 04:30 PM	Student Visible
Contact Info	Institutional Email, Phone Number, Recovery Details	Restricted Admin

5. Timetable

Master Institutional Timetable Architecture

The Timetable module operates on a Master Timetable Architecture. Academic Management defines one central institutional timetable, from which every faculty member's personal schedule is dynamically derived.

Timetable Features

- **Master Synchronization:** Modifications in Academic Management instantly update Faculty Timetables and Dashboard Widgets.
- **6 View Switchers:** Supports Institution View, Department View, Faculty View, Room View, Laboratory View, and Section View.
- **Conflict Validation:** Automated detection of faculty double booking, classroom overlaps, and section schedule conflicts.
- **Section Timetable Drawer:** One-click drawer overlay displaying detailed section weekly rosters and course distribution.

6. Subjects

Course Allocation & Curriculum Mapping

Lists all assigned theory and laboratory courses allocated to the faculty for the active semester, complete with syllabus progress, credits, and section rosters.

Course Code	Course Title	Credits	Type	Assigned Section
CS601	Operating Systems	4.0	Theory	CSE-A / CSE-B
CS601L	Operating Systems Laboratory	2.0	Lab	CSE-A (Batch 1 & 2)
CS602	Database Management Systems	4.0	Theory	CSE-A
CS603	Computer Networks	3.0	Theory	CSE-B

7. Lesson Plans

Session Planner & Syllabus Tracking

Provides a granular breakdown of course syllabus completion, planned teaching hours, covered topics, and course outcome (CO) attainment tracking.

Planner Components

- **Teaching Hour Cards:** Compares Total Allocated Hours vs. Conducted Lectures vs. Remaining Syllabus Hours.
- **Covered vs. Remaining Topics:** Interactive checkboxes tracking unit-wise topic completion percentages.
- **Teaching Resources:** Direct attachment of lecture slides, textbook chapter references, and lab manuals to specific sessions.
- **Weekly & Monthly Timeline:** Visual progress bar showing pace of course delivery against academic calendar milestones.

8. Attendance

Daily Section Roster & Defaulter Tracking

Allows faculty to record daily student attendance in under 30 seconds using an interactive student grid or list view with automatic defaulter detection.

Key Functionality

- **Quick Mark & Bulk Actions:** One-click 'Mark All Present' with toggle options for Absent or Late status.
- **Defaulter Threshold Alerts:** Automatic red badge highlighting students falling below the mandatory 75% attendance threshold.
- **Export Options:** Generates instant monthly attendance logs in Excel (.xlsx) and CSV formats for department records.

9. Students

Mentoring & Academic Performance Monitoring

Enables faculty to view student profiles, monitor academic performance, track internal assessment marks, and handle mentoring queries.

- **Student Roster View:** Comprehensive list of enrolled students with roll numbers, registration IDs, and photo avatars.
- **Performance Analytics:** Historical GPA, semester attendance records, and backlogs overview.
- **Direct Messaging:** In-app communication channel for answering academic queries and posting course updates.

10. Assignments

Digital Task Allocation & Online Grading

Streamlines creation, distribution, submission tracking, and grading of student course assignments.

Assignment Title	Course	Due Date	Max Marks	Submissions Status
Process Synchronization & Semaphores	CS601	15 Aug 2026	20	58 / 64 Submitted
SQL Complex Queries & Joins	CS602	20 Aug 2026	15	62 / 64 Submitted
Socket Programming in C	CS603	25 Aug 2026	25	45 / 60 Submitted

11. Study Materials

Centralized Digital Resource Library

Serves as the digital repository for course slides, lecture notes, reference textbooks, lab manuals, and previous year question papers.

Module Highlights

- **Multi-Format Uploads:** Supports PDF documents, PowerPoint slides (.pptx), Word notes (.docx), and ZIP lab archives.
- **Role-Based Visibility:** Restrict material access by Department, Semester, Section, or Public Public Access.
- **Download Analytics:** Tracks student download metrics to evaluate material engagement.

12. Assessments

Continuous Internal Evaluation (CIE)

Manages internal tests, mid-semester exams, quizzes, lab practical vivas, and mini-project evaluations with automatic grade computation.

Assessment Name	Type	Weightage	Max Marks	Publishing Status
Mid-Semester Internal Exam 1	Written Exam	20%	50	Published
Database Design Project	Project	10%	25	Marks Pending
Linux Shell Scripting Lab Viva	Practical	10%	25	Published

13. Examinations

Semester Examinations & Invigilation Duty Center

Standardized dashboard for invigilation duty rosters, exam hall seat allocations, question paper submissions, and semester end evaluation.

Key Features

- **Invigilation Duty Schedule:** Displays assigned exam blocks, date, hall number, and session timing.
- **Question Paper Repository:** Secure encrypted upload portal for mid-term and semester end question papers.
- **Seating Plan Viewer:** View hall-wise student roll number seating matrices.

14. Research

Faculty Publications, Patents & Grants Portfolio

Dedicated research workspace tracking Scopus/SCI indexed journal publications, international conference proceedings, patents, research grants, and consultancy projects.

Category	Metrics Count	Primary Indexing / Focus
Journal Publications	14 Papers	Scopus / SCI / SCIE Indexed
Conference Papers	8 Papers	IEEE / ACM / Springer
Ongoing Projects	3 Grants	DST / AICTE Sponsored (₹45 Lakhs)
Patents Granted	2 Patents	Indian Patent Office (IPO)

15. Leave

Leave Applications & Approval Workflow

Enables faculty to apply for casual leave (CL), earned leave (EL), medical leave (ML), and on-official-duty (OOD) with automated HOD approval routing.

- **Leave Balances:** Real-time summary cards displaying remaining leave quotas per category.
- **Class Alternate Arrangement:** Mandatory selection of substitute faculty for scheduled lectures during leave.
- **Approval History:** Complete audit trail of submitted, approved, and rejected leave applications.

16. Payroll

Salary Statements & Tax Downloads

Provides secure access to monthly payslips, salary breakdowns, allowances (HRA, DA, Special Allowance), tax deductions (TDS, PF), and Form 16.

Monthly Salary Component Structure

- Basic Pay: Scale based on Academic Level / Designation
- Dearness Allowance (DA): Institutional percentage rate
- House Rent Allowance (HRA): Metro / Regional rate
- Deductions: Provident Fund (PF), Professional Tax (PT), Income Tax (TDS)
- Downloads: One-click monthly payslip PDF and Form 16 export

17. Reports

Institutional Compliance & Accreditation Reports

Automates generation of official reports required for internal academic audits, university submissions, and accreditation bodies (NBA / NAAC).

Report Types

- **Attendance Deficiency Reports:** Lists students below 75% attendance for condonation or hall ticket withholding.
- **Internal Assessment Reports:** Consolidated marks sheets ready for university portal upload.
- **NBA OBE Reports:** Course Outcome (CO) - Program Outcome (PO) attainment calculation sheets.
- **NAAC Criteria Reports:** Criteria 2 (Teaching-Learning), Criteria 3 (Research & Extension), and Criteria 5 (Student Support).

18. Notifications

ERP Notification Center

Centralized notification center keeping faculty updated on critical academic deadlines, student leave requests, research updates, and institutional circulars.

- **4 KPI Summary Cards:** Unread Notifications, High Priority, Today's Notifications, Action Required.
- **11 Category Filters:** Filter by Assignments, Attendance, Examinations, Leave, Research, Students, System, Announcements, Timetable.
- **Direct Context Actions:** [View Assignment], [Take Attendance], [Enter Marks], [Review Leave Request].

19. Settings

Account & Personalization Center

Redesigned settings center providing complete control over account parameters, security credentials, notification channels, and teaching defaults.

Settings Section	Configurable Parameters	Actions
Profile Settings	Name, Designation, Email, Phone, Cabin, Office Hours	Edit Profile
Account Security	Change Password, 2FA Toggle, Active Sessions List	Logout Devices
Notification Prefs	Email Notifications, Push Notifications, Category Toggles	Toggle Switches
Appearance	Theme (Light/Dark/System), Compact Mode, Language	Preferences
Privacy	Profile Visibility, Hide Email, Hide Phone Number	Privacy Scope
Teaching Prefs	Default Semester, Department, Section, Attendance View	Save Defaults
Download Center	Export CV, Activity Log, Teaching History, Research PDF	Download PDF
Danger Zone	Deactivate Account, Logout All Devices, Delete Request	Warning Actions

20. Role-Based Access Control

Security Matrix & Scope Boundaries

The system enforces strict Role-Based Access Control (RBAC) to ensure data confidentiality and multi-tenant security across departments.

- **Department Awareness:** Faculty members can only access student rosters and courses belonging to their assigned department.
- **Semester & Section Isolation:** Marks and attendance entry are strictly restricted to assigned course sections.
- **Student Data Protection:** Sensitive personal information (phone numbers, salary details, financial records) are hidden from unauthorized users.

21. Future Backend APIs Specification

RESTful API Endpoints Contract

The frontend architecture is fully prepared for backend integration. Below is the complete API specification table for all Faculty Module endpoints:

HTTP Method	API Endpoint Route	Description / Purpose
GET	/api/v1/faculty/profile	Fetch logged-in faculty profile details
PUT	/api/v1/faculty/profile	Update profile details & office hours
GET	/api/v1/timetable/master	Fetch centralized master timetable
GET	/api/v1/timetable/faculty/:id	Fetch derived personal faculty timetable
POST	/api/v1/attendance/mark	Submit section attendance roster
GET	/api/v1/lesson-plans	Fetch course session planner progress
POST	/api/v1/materials/upload	Upload new study material / PDF
POST	/api/v1/assignments/create	Create new student assignment
PUT	/api/v1/assignments/:id/evaluate	Submit student assignment marks
POST	/api/v1/assessments/marks	Submit continuous evaluation marks
POST	/api/v1/research/publication	Add new research journal publication
POST	/api/v1/leave/apply	Submit leave application with substitute
GET	/api/v1/payroll/payslips	Fetch monthly salary payslip PDF
GET	/api/v1/notifications	Fetch notification list with filters
PUT	/api/v1/settings	Save user preferences & security config

22. Database Entities & Schema Models

Enterprise Entity Relationship Specification

The core database schema models defined for the backend implementation of the Faculty Module include:

Core Database Schema Entities

- Faculty: id, employeeId, name, email, phone, designation, departmentId, cabin, photoUrl
- Department: id, deptCode, name, hodFacultyId, building
- Subject: id, code, title, credits, semester, departmentId, lectureType
- TimetableEntry: id, academicYear, regulation, deptCode, semester, section, subjectId, facultyId, room, day, period
- AttendanceRecord: id, date, sectionId, subjectId, facultyId, studentId, status ('Present'|'Absent'|'Late')
- LessonPlan: id, subjectId, unitNumber, topicTitle, plannedHours, status ('Covered'|'Pending')
- StudyMaterial: id, title, fileUrl, fileType, departmentId, semester, uploadedByFacultyId
- Assignment: id, title, description, dueDate, maxMarks, subjectId, createdByFacultyId
- AssessmentMark: id, assessmentId, studentId, marksObtained, maxMarks, evaluatorFacultyId
- ResearchPublication: id, facultyId, title, journal, doi, impactFactor, indexedIn ('Scopus'|'SCI')
- LeaveRequest: id, facultyId, leaveType, startDate, endDate, substituteFacultyId, status
- PayrollRecord: id, facultyId, month, year, basicPay, hra, da, deductions, netPay
- Notification: id, userId, title, message, category, priority, isRead, createdAt
- UserSettings: id, facultyId, theme, fontSize, emailAlerts, pushAlerts, defaultSemester

23. UI Screens Specification

Frontend Component Architecture Mapping

Page Route	Page Title / View Name	Primary Components Used
/faculty	Faculty Dashboard	TodaySchedule, AllInsights, QuickActions, KPIWidget
/faculty/profile	My Profile	ProfileHeader, AcademicDetailsCard, ResearchCard
/faculty/timetable	Timetable & Schedule	WeeklyGridMaster, SectionDrawer, FilterPanel
/faculty/subjects	My Subjects	SubjectCardList, SyllabusProgressBar, CourseRoster
/faculty/lesson-plan	Lesson Planning	SessionPlanner, TeachingHoursCard, TopicList
/faculty/materials	Study Materials	UploadModal, SubjectAccordion, VisibilityPanel
/faculty/notifications	Notifications Center	NotificationHeader, NotificationCard, Filters
/faculty/settings	Account Settings	ProfileCard, SecurityCard, SaveBar, DangerZone

24. Technology Stack

Modern Frontend & Future Backend Stack

Tier Layer	Technology Framework	Purpose & Role
Frontend UI	React 19, TypeScript, Tailwind CSS v4	High-performance reactive user interface
UI Components	ShadCN UI, Radix Primitives, Lucide Icons	Accessible, styled UI design system
Routing & State	TanStack Router, React Context API	Client-side routing & state orchestration
Data Visualization	Recharts Data Library	Responsive KPI charts & donut visualizations
Future Backend	Node.js, Express.js, TypeScript	Scalable RESTful API server
Database	MongoDB / PostgreSQL	Enterprise relational & document storage
Authentication	JWT, OAuth 2.0, bcrypt hashing	Secure session & token management

25. Security & Authentication

Data Protection & Privacy Architecture

Security is embedded into every layer of the Faculty Module, ensuring compliance with institutional privacy standards and ISO/IEC 27001 data protection guidelines:

- **Authentication:** JSON Web Tokens (JWT) stored in HTTP-only secure cookies with 24-hour expiration.
- **Multi-Factor Auth (MFA):** Optional TOTP authenticator app support for administrative & HOD actions.
- **Authorization:** Granular Role-Based Access Control (RBAC) middleware verifying permissions on every route.
- **Data Privacy:** Personal faculty contact information and financial payslips are encrypted at rest and in transit (TLS 1.3).

26. Responsive Design

Cross-Platform Device Compatibility

The Faculty Module features a fully responsive layout engineered to deliver a seamless experience across all device form factors:

Device Type	Breakpoint Range	Adapted Layout Behavior
Desktop Monitors	"e 1280px (xl / 2xl)	Full multi-column dashboard, side-by-side matrices & floating bars
Laptops & Tablets	768px - 1024px (md / lg)	Collapsible sidebar, horizontal scrollable grids & stacked cards
Mobile Smartphones	< 768px (sm)	Day-wise card views, bottom drawer sheets, full-width touch targets

27. Future Enhancements

AI-Powered Academic Tools & Expansion Roadmap

Planned future enhancements to further elevate the ERP experience for faculty include:

- **AI Teaching Assistant:** Intelligent chatbot assistant recommending lecture exercises and answering student queries.
- **Automated Lesson Planner:** AI-driven curriculum ingestion generating 16-week lecture schedules automatically.
- **Attendance Predictive Analytics:** Machine learning model predicting student dropout risks based on early attendance trends.
- **PDF OCR Ingestion:** Automated digitizer converting scanned question papers and notes into searchable LMS resources.

28. Conclusion

System Summary & Readiness Statement

The Faculty Module of the EduSuite Pro College Management System provides a modern, intuitive, and robust ERP platform built specifically for faculty productivity. With its single-source Master Timetable Architecture, department-aware data scoping, continuous assessment tools, and backend-ready API contract, the module establishes a state-of-the-art foundation for modern higher education institutions.

System Readiness & Approval Summary

- ' Master Timetable Architecture: Fully Synchronized & Operational
- ' Notifications Center: Centralized & Category Filtered
- ' Account & Settings Center: 8 Dedicated Sections & Floating Save Bar
- ' UI / UX Design: ERP Primary Blue Theme, Responsive & Accessible
- ' API Contract: Fully Specified & Ready for Backend Integration
- ' Final Recommendation: Approved for Deployment & Production Use

